1 Powers of Tangent and Cotangent

Let't first look at even powers of tangent and cotangent.

1. Evaluate $\int \tan^2(x)dx$. (*Hint*: Use the identity $1+\tan^2(x) = \sec^2(x)$.)

2. Evaluate $\int \tan^4(x)dx$.

3. Evaluate $\int \tan^6(x)dx$.

4. Based on your previous answers, make a conjecture for $\int \tan^{2n}(x)dx$ when $n \in \mathbb{N}$. Use your conjecture to evaluate $\int \tan^{10}(x)dx$ without doing additional work.

One can work out a similar formula for $\int \cot^{2n}(x)dx$ when $n \in \mathbb{N}$, by using the identity $1+\cot^2(x) = \csc^2(x)$. The details are left to the interested reader, as are the proofs of these formulas. Instead, we now move on to odd powers of tangent and cotangent. These are also computed recursively, using the same trig identities.

We have proved previously that

$$\int \tan(x) = \ln|\sec(x)| + C$$
$$\int \cot(x) = \ln|\sin(x)| + C.$$

5. Evaluate $\int \tan^3(x)dx$.

6. Evaluate $\int \tan^5(x)dx$.

7. Based on your previous answers, make a conjecture for $\int \tan^{2n+1}(x)dx$ when $n \in \mathbb{N}$. Use your conjecture to evaluate $\int \tan^9(x)dx$ without doing additional work.

Again, similar formulas hold for cotangent. The details are left to the interested reader.

2 $\int \tan^m(x)\sec^n(x)dx$

In the previous section, we encountered integrals of the form $\int \tan^m(x)\sec^2(x)dx$, where m is a positive integer. These can all be evaluated by letting $u = \tan(x)$, which gives

$$\int \tan^m(x)\sec^2(x)dx = \int u^m du = \frac{\tan^{m+1}(x)}{m+1} + C, \quad m \in \mathbb{N}.$$

We now consider other cases. First, suppose the power of secant is even.

8. Evaluate $\int \tan^m(x)\sec^4(x)dx$ when $m \in \mathbb{N}$.

9. Explain in your own words how to evaluate $\int \tan^m(x)\sec^{2n}(x)dx$ when $m, n \in \mathbb{N}$.

Now suppose the power of tangent is odd.

10. Evaluate $\int \tan(x)\sec^n(x)dx$.

11. Evaluate $\int \tan^3(x)\sec^n(x)dx$.

12. Explain in your own words how to evaluate $\int \tan^{2m+1}(x)\sec^n(x)dx$ when $m,n \in \mathbb{N}$.

For other cases, the guidelines are not as clear-cut. We may need to use identities, Integration by Parts, and occasionally a little ingenuity.

13. Evaluate $\int \tan^2(x)\sec(x)dx$.

A similar analysis can be carried out for integrals of the form $\int \cot^m(x)\csc^n(x)dx$ as well as for powers of secant and cosecant.